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SARDAR PATEL UNIVERSITY
BCA Semester-6 Examination, 2021
US06CBCA02 || Computer Networks [NC]
16th July, 2021.

Time : 10.00 A.m. – 12.00 p.m.

Marks: 70

Q-1

Select the appropriate option for the following questions:

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- 1) "Interconnected collection of autonomous computers" is known as _____.
A. Computer network C. Host
B. Server D. Subnet
- 2) The connection of two or more networks is called an _____.
A. computer network C. internet
B. internetwork D. local Area Network
- 3) _____ cable consists of an inner copper core and a second conducting outer sheath.
A. Twisted-pair C. Coaxial
B. Fiber-optic D. Shielded twisted-pair
- 4) Transmission media are usually categorized as _____.
A. Fixed or unfixed C. Determinate or indeterminate
B. Guided or unguided D. Metallic or nonmetallic
- 5) Dialog control and synchronization are responsibilities of _____.
A. Data link layer C. Transport layer
B. Network layer D. Session layer
- 6) An artificial satellite needs to have _____ which is the path in which it travels around the Earth.
A. Channel C. Footprint
B. Tower D. Orbit
- 7) Which of the following CSMA has highest chances of collision?
A. 1-persistent C. p-persistent
B. non-persistent D. All have equal chances
- 8) A repeater operates at _____ layer.
A. Physical C. Datalink
B. Network D. Transport
- 9) For long distance data transmission, what is the preferable mode of communication?
A. Serial transmission C. Half duplex
B. Parallel transmission D. Full duplex
- 10) When the packets are small and all the same size, they are often called _____.
A. Cells C. Trunks
B. Circuits D. Subnet

(1)

(P.T.O.)

Q-2 Tick (TRUE or FALSE) and Fill in the blanks.

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- 1) DQDB stands for Double Queue Double Bus. [TRUE/FALSE]
- 2) When a beam of light travels through media of two different densities, if the angle of incidence is greater than the critical angle, Reflection occurs. [TRUE/FALSE]
- 3) OSI stands for Open Standard Interconnection. [TRUE/FALSE]
- 4) A router operates at Network layer. [TRUE/FALSE]
- 5) _____ network consists of many connection between individual parts of machines.
- 6) Transmission media are usually categorized as _____.
- 7) The signal from a satellite is normally aimed at a specific area called _____.
- 8) CSMA stand for _____.

Q-3 Give short answers of the following questions: (ANY TEN)

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- 1) Explain DQDB in brief.
- 2) Define terms: Data Rate, Server
- 3) What are the advantages of computer network?
- 4) Explain application of twisted pair in brief.
- 5) List various transmission modes.
- 6) Write the difference between circuit switching and packet switching.
- 7) Define: Protocol Stack and Network architecture.
- 8) List categories of satellite.
- 9) List responsibilities of Physical layer of OSI model.
- 10) Explain in brief: Broadcast network.
- 11) Draw the diagram of Mesh topology.
- 12) Explain the function of modem.

Q-4 Give Long answers of the following questions: (ANY FOUR)

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- 1) Explain WAN in detail.
- 2) How can we classify computer Network? Explain.
- 3) Explain Coaxial cable in detail.
- 4) Explain FDM in detail.
- 5) What is OSI? List all layers of OSI model. Explain each layer in few lines.
- 6) List the types of communication satellite. Explain geosynchronous satellite.
- 7) Write a short note on : Repeater and Bridge
- 8) Explain Star topology and Bus topology with their advantages & disadvantages.

—X—

(2)